Artificial Intelligence

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“Large Language Model” (LLM) Demo

- Write and Revise a Story (Claude Sonnet)
  - Understands the context of the story
  - Summarizes; changes style
- Multimodal: ChatGPT cat photo
- Many LLMs available, steadily improving
  - ChatGPT (OpenAI), Claude (Anthropic), R1 (DeepSeek)

Using an LLM

- Typical uses
  - Comprehensive web search
  - Summarizing and analyzing documents (Emancipation Proclamation example)
  - Generating or editing prose (passive voice change)
  - Tutorial discussions
  - Brainstorming and recommendations (AZ cities visit)
  - Language translation
- Learning to Prompt
  - More than a question. Note how ChatGPT rephrases
  - Prompt and entire conversation are “input” for further conversation

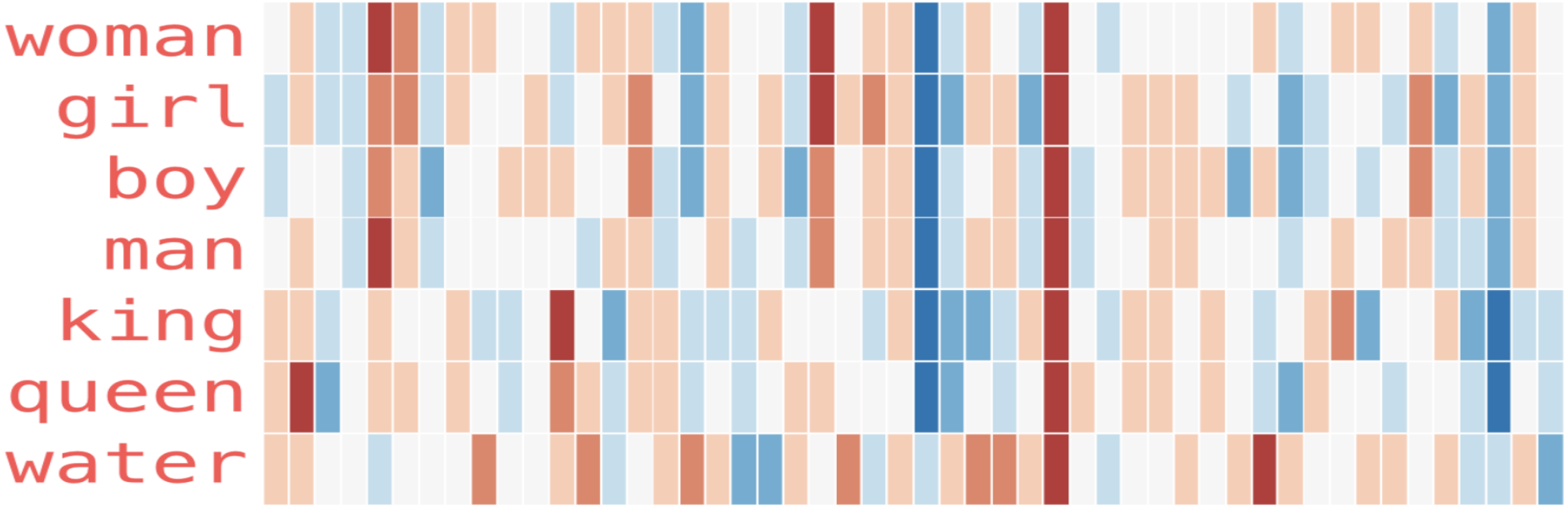
An LLM Is *Not*

- An improved search engine
  - Is able to web-search when useful, but…
  - … can work unplugged from the internet
- Hand-crafted code
  - No programming specific to the questions/answers
  - Could be written in one page of code – **12 equations**
- The whole modern AI story
  - Important other areas are making major advances too, e.g. “reinforcement learning”, computer vision

An LLM *Is*

- A sophisticated mathematical engine
  - 12 equations, but on “matrices” – huge grids of numbers
  - From 10 billion to over 1 trillion numbers
  - The “intelligence” is in those numbers (aka weights or parameters)
- Self-organizing!
  - The weights are “trained” by incremental adjustment to match known texts and human feedback.
  - Humans design the training process and the equations, but not the weights

Word/Idea “Embedding”



- This is how an LLM represents words, phrases and ideas
- Each word has a number sequence – an “embedding”
- But, with 12,800 numbers per word – many conceptual dimensions

How An LLM “Thinks”

- Starts with word embeddings representing words in your prompt and the current conversation.
- Combines words into higher 12,800-number patterns representing phrases, then sentences, etc.
- Predicts next “thought” as a number pattern and chooses closest next word.
- Repeat, word by word...

The Weights *are* the “Intelligence”

- All “reasoning” is by word/concept embeddings.
- And we (mostly) *don’t know* what each weight does.
- Could learn to converse in a language none of its programmers understand.
- Are world-models, theory of mind, etc. spontaneously emerging from the training?

Is an LLM Intelligent?

- “Yes” arguments
  - Vast possibilities in 1 trillion weights. Printed out they’d be 4 million large books.
  - Behavior suggests world-models, theory of mind, etc.
  - (But, we really don’t know since we’re not sure how the weights work in detail)
- “No” arguments
  - Just mindless numbers and 12 equations
  - Just predicting the next word.
  - (But, it’s improving *fast*, a human brain is just “mindless” neurons and synapses, and *how* is it picking that next word so intelligently?)

“Other” AI

- Diffusion Models
  - DALL-E example (Mammoths on the chapel green, Sora video)
- Reinforcement Learning
  - Self-instructing, based on automatically generable feedback
  - Alpha Go example
- Computer Vision
  - CNNs and Vision Transformers
  - Recognize and interpret images and visual data (Cat humor)